

Project 1, Session 4: Data visualization I: visual style

- Paired team review
- Style for legible and interpretable figures
- Stylistic consistency and meaning
- Boolean indexing and merging dataframes

Team discussion

Paired teams: 5-7-9, 1-6, 2-8, 3-4

- How did you compute coverage (number of observed missense variants) across all PTEN positions?
- How did you compute score statistics for each position, with a filter for the `minimum_variants` for inclusion?
- Which statistics did you choose to compute and why?
- How did you verify that your code to compute abundance and score statistics performs as expected and is robust?
- How did you visualize coverage and score statistics across the PTEN protein sequence, and what patterns did you observe?

When presenting, make sure to **clearly describe your code** and what it does.

Coverage and median abundance along the PTEN sequence

Which PTEN positions have lower median abundance, and how much data supports each point? We can visualize these quantities across the sequence.

```
show_plot(s4_overview)
```

Coverage counts substitutions. Medians summarize eligible positions.

Visual style guidelines for figures

When composing figures to share (even informally!), generally we want to ensure that others can understand what we are plotting and how to interpret it.

Labels should be

- descriptive
- legible

Reference scales should be clearly indicated, when they exist.

Color and shape should carry consistent meaning.

Figures are a form of **communication**. We should create them with **intention**.

Labels carry scientific meaning

Could the left figure be interpreted without additional context?

```
show_plot(s4_labels)
```

Include units in labels when relevant.

Note the WT reference. What is a "high" abundance score and what is a "low" one? These are anchored relative to WT.

Background gridlines can serve as a reference but should not overwhelm the data.

Figure size and readable text

Both figures use 12-point text. Why does the right-hand text look smaller here?

```
show_size_demo()
```

Both images occupy the same display width. In the matplotlib package, `figsize` sets the dimensions of the figure. Choose dimensions equal to the final presentation size, which would be different for a figure in a paper, a presentation slide, or a poster.

Color (and shape) can be used to consistently convey meaning

We naturally recognize patterns in color and shape. It can be confusing if their meaning changes between figures or between different panels of a multi-part figure. Choosing a consistent theme is one way to make your results easier to understand.

```
show_plot(s4_category_colors) # artificial scores for illustration
```

Select distinct categorical colors for groups without a clear numerical order.

Note: color blindness is very common (~5% of population)! Tools can help select color schemes that are easy to interpret (e.g., <https://colorbrewer2.org/> by Cynthia Brewer).

Two ways of encoding quantitative information in color

Does the quantity have an **ordering**, or a meaningful reference with **two sides**?

Such quantities are commonly represented by **sequential** and **diverging** color palettes, respectively.

```
show_plot(s4_palette_types)
```

Sequential: ordered amount. **Diverging:** departure from a reference.

Colorbars should include a label and visible scale marks.

Some sequential color schemes can create the artificial appearance of sharp boundaries

For the left color scheme, small changes in the median abundance can sometimes lead to large changes in the perceived color. The right color scheme is "perceptually uniform", such that equal changes in abundance should produce correspondingly equal changes in perceived color.

```
show_plot(s4_artificial_edges)
```

Avoid creating the appearance of sharp boundaries between nearby values unless supported scientifically.

Diverging color schemes need a meaningful reference point

On the left, does a warm color mean "above the WT reference"?

```
show_plot(s4_meaningful_center)
```

For PTEN abundance a natural reference is 1 (WT). For a difference it may be 0. In general, the appropriate reference depends on the question.

Equal departures and color intensity

Compare 0.75 and 1.25: both are 0.25 from the reference. Should they have similar intensity?

```
show_plot(s4_normalization_choices)
```

The center fixes the reference and numerical spans fix how quickly color changes. A different span could be appropriate if scales on either side of the reference should be interpreted differently.

Selecting and summarizing data in pandas

Pandas has useful tools for selecting subsets of data, summarizing, and comparing them.

Let's examine a toy dataset of variant scores for an artificial protein with four positions. `expts` describes the number of experimental replicates in which the variant was observed.

```
variants
```

Split, calculate, combine

Collect rows with the same position, then summarize their scores.

Prediction sheet A: For position 1, what is the count, median, and mean score?

```
grouped = variants.groupby("position")["score"]  
grouped.agg(["count", "median", "mean"])
```

Filter first; then recount coverage

What happens if we only consider variants observed in at least six experiments?

Prediction sheet B: What will the count and median score be at position 1 after applying this filter? If we require at least 5 variants to be measured at a position to report a score, could we still do so for position 1?

```
retained = variants.loc[variants["expts"] >= 6]
retained.groupby("position")["score"].agg(["count",
"median"])
```

Boolean selection

More generally, we can select any subset of a dataframe by selecting rows based on any boolean condition.

```
keep = variants["expts"] >= 6  
keep
```

```
variants['keep'] = keep
```

```
# both expressions yield the same view of the variants  
dataframe  
variants.loc[variants['keep']]  
variants[variants['keep']]
```

Comparing two dataframes

Prediction sheet C: Which position has two comparable values?

What would subtracting the second rows mean?

```
import pandas as pd

baseline = pd.DataFrame({"position": [1, 2],
                          "median_score": [0.4, 0.8]})
alternative = pd.DataFrame({"position": [1, 3],
                            "median_score": [0.5, 0.9]})
display(baseline, alternative)
```

When comparing across dataframes, we want to ensure that rows are matched by a suitable **identifier**, not just row order in the dataframe.

Keep every position in either table

A **key** identifies the object being matched. Here we **merge** dataframes with `position` as the key.

Columns that have the same name in `baseline` and `alternative` are given new names with suffixes `_baseline` and `_alternative`, respectively.

Here, `validate="one_to_one"` is an option that will reject duplicate position keys.

```
matched = baseline.merge(  
    alternative, on="position", how="outer",  
    suffixes=("_baseline", "_alternative"),  
    indicator=True, validate="one_to_one",  
)  
matched
```

The `_merge` column describes whether the key is in both dataframes, the "left" one only (`baseline`) or the "right" one only (`alternative`).

Missing comparison is different from zero change

Prediction sheet D: Which differences are defined, and what is their sign?

```
matched["difference"] = (  
    matched["median_score_alternative"] -  
    matched["median_score_baseline"]  
)  
matched[["position", "difference", "_merge"]]
```

Assignment for next time

With your team, we will associate different parts of PTEN with different functional regions and test how sensitive our results are to the filtering criteria we've defined.

- Assign each variant a **region** based on the supplied annotations.
- For each region, summarize the number of included variants, number of included sites, and total number of sites in the region.
- For each region, compute and interpret 1) the median of all variant scores in the region, and 2) the median of the median score at each site in the region, across sites.
- Visualize score statistics by region.
- Test the sensitivity of the results: what happens if you change `minimum_variants` required for a site to be included, or the minimum number of experiments that a variant must be measured in?

For the beginning of class, be ready to

- Interpret boolean selection for dataframes

- Compare values across merged dataframes